From a Cold Clone to a Proof Receipt

Agent-native navigation and incremental validation for a large Lean development

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Abstract

A large formal library can be mechanically exact and still be practically unreadable to the next human or reasoning agent. This paper presents a repository architecture that separates first-contact comprehension from proof checking. At the audited revision, a cold clone exposes a public proof cockpit that composes checkout and toolchain identity, corpus scale, exact open propositions, problem obligations, and committed workbench receipts. Beneath that card sit a bounded tour card, a mathematical map of ten thematic programmes across eight indexed Erdős problems, a reviewed public claim map with explicit open propositions, and an elaborated loaded-root reference graph joined to source coordinates. These are committed navigation products: they require no Lean build and make omissions and authority boundaries explicit. The inventory behind them spans 1,019 Lean modules and 153,253 declarations; 503 of those modules and 8,171 of those declarations are explicitly marked machine-generated certificate shards, counted as formal source and never as separate mathematical claims. The marked figures are a classification floor, not the generated share: large emitted families predate the markers, so the true machine-generated share is substantially higher.

Navigation does not receive proof authority. A session notary records an agent's observations, falsifiable conjectures, abandoned routes, exact Lean probes, and claims. The notary derives probe verdicts from the pinned Lean process; an agent cannot supply them through its interface. Claims must cite an accepted probe, and replay reruns the stored bytes. Compilation is similarly separated from orientation. Lake outputs and content traces support focused or changed-cone builds, while an exact cached receipt prevents an unchanged dependency-index check from repeating a full environment export. A recorded case study contains six reasoning notes, one accepted probe, and two claims, and its probe replays at the audited revision.

The architecture combines five functions without confusing their roles: complete inventory, selective interpretation, short intent-based routes, recorded reasoning, and incremental validation. It is not an autonomous theorem prover, a portability study, or evidence that the agent's reasoning was optimal or mathematically novel outside the repository history. It is a prototype audited in the author's environment. As of 14 September 2026, no completed external cold-clone use had been recorded.

Authorship and AI use. Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

What this paper establishes

Result. A cold clone exposes the library's complete declaration inventory, reviewed mathematical routes, public claim boundaries, and exact source links before Lean is built. The notary derives accepted proof-probe verdicts from pinned Lean execution and can replay the stored bytes. **Evidence.** The audited repository contains 1,019 modules and 153,253 declarations; the paper includes a replayed session case study. **Limit.** This is a single-environment prototype with no recorded external cold-clone use.

1 The cold-clone problem

The first interaction with a large Lean repository is usually a filesystem. The reader sees source files, generated files, scripts, papers, and build configuration, but not the shape of the mathematics. A language model can search filenames and text, yet those operations do not answer the questions that matter: Which programmes exist? Which propositions are established, conditional, finite, or open? Where does a declaration sit in the formal dependency graph? Which surface is exhaustive, and which is a human selection? What may be trusted without compiling anything?

Poor first-contact navigation changes the mathematics an agent attempts. An agent that cannot see a library's option surface may rediscover existing lemmas, confuse a finite result with its open unbounded neighbour, or build a new local index because the existing one was not discoverable. Loading every declaration, paper, and proof edge into one prompt destroys the distinctions the extra context was meant to reveal. A useful first-contact surface must therefore be small, expandable, and honest about what it omits.

The development studied here indexes eight Erdős problem programmes. The ten map entries are thematic routes across these problems, not ten separately indexed problems. The claim registry now covers selected statements from all eight; a paper's inclusion does not give every statement in it formal status. The degree-seven counterexample due to the erdosproblems.com contributor `ani`, formalised in Lean in this repository, refutes the Formal Conjectures 1041 one-dimensional Hausdorff path-image claim, while correspondence with its historical curve-length question remains unreviewed and the other seven targets remain unresolved here. The detailed navigation examples in this paper concentrate on Problems 249 and 257: exact separation equivalences for Problem 249, a machine-checked rational countermodel agreeing with the totient's parity at every index, and quotient-greedy classifications for Problem 257, with obstruction theorems recorded beside the routes they close. The generated certificate shards named in the abstract sit underneath these results as checked finite evidence, not beside them as further claims. The design goal is not to replace reasoning with a fixed pipeline. It is to let a capable reader see enough structure to reason well, while reserving mathematical authority for the pinned proof kernel.

The paper first separates the repository's four navigation layers and then shows how a bounded tour expands into exact source and dependency routes. Sections 4 and 5 mark the transition from navigation to kernel authority and explain the incremental validation path. Section 6 gives a replayed case study; the remaining sections compare related systems and state the limits of transfer.

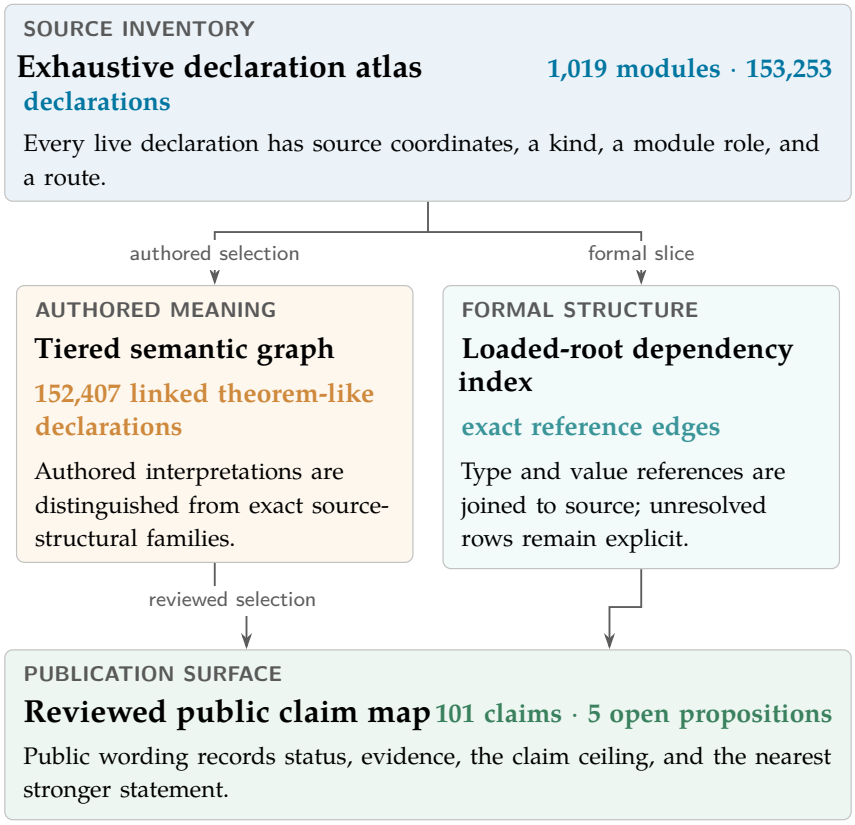


Figure 1: The navigation layers. The dependency index and semantic graph are different partial views of the exhaustive atlas; neither is presented as a complete interpretation.

This paper owns the navigation and validation case study. The wider claim-transition lifecycle is described in the systems paper, while the contribution protocol specifies how an outsider returns a change. These links lead to named destinations in the sibling PDFs rather than to page numbers.

The same proof-state compiler also supports bounded inspection of a selected elaborated proof. The existing feedback experiment uses that route to expose admissible construction choices and emits finite obligations for the same workbench to replay. Comparator can check the separately stated continuation contract through a named unit of its existing runner. The implementation, literature attribution and finite controls are described in the systems paper's discussion of executable construction choices. These operations extend the existing research route and do not automatically extract a new contract or establish an infinite claim from a finite run.

2 Four views of the mathematical library

No single graph can serve all readers without hiding an important distinction. The repository therefore commits four related layers (Figure 1). They are projections with different coverage contracts, not successive claims of understanding.

Complete inventory. The declaration atlas is a source-derived address book. It records every live declaration that its explicit source roots contain. Exhaustiveness here means that a reader can locate a declaration, not that the system understands its mathematical role.

Exact formal neighbourhoods. The dependency index loads the two supported compact roots, extracts direct constant references from elaborated types and values, and joins those constants to atlas coordinates. It reports unresolved rows and edges instead of silently treating absence as independence. Its graph is exact for the loaded environment and stated edge relation, but it is not a transitive proof explanation.

Authored mathematical interpretation. The semantic graph contains authored statement nodes, typed relations, and an exact source-structural floor. At the audited revision all 152,407 theorem-like declarations in the source scope are linked. Of these, 139,818 (91.7%) participate in authored mathematical interpretations: 3,330 are exact proposition evidence and 136,488 are bounded contextual links to digest- or module-verified families. The remaining 12,589 are grouped only by exact source module and normalised Lean proposition signature. That lower tier is useful navigation, not a mathematical paraphrase. Keeping the tiers visible prevents exhaustive linkage or bulk helper assignment from being misreported as exhaustive direct understanding. A paper-seeded population query continues to rank exact live citations whose best route is structural rather than authored. The declaration count does not measure distinct discoveries, human-authored proofs, or individually reviewed mathematical interpretations.

Reviewed public claims. At the audited revision the claim map selected 101 results for public exposition. It recorded 37 as proved here, 8 as formalised here, 5 as unconditional progress, 39 as conditional reductions, 7 as verified finite instances, 3 as cited only, and 2 as open. Five explicit frontier propositions described the stronger obligations that survived; the current registry is larger. These labels are maintainer-reviewed public meaning, not outputs inferred by the proof kernel.

3 A short tour with exact follow-up queries

The public entry command is `python3 scripts/proof_cockpit.py`. Its compact card combines the checkout, pinned toolchain, corpus scale, exact open frontier, problem-local obligations, recorded workbench outcomes, and the next typed commands. Every field comes from the public clone. No private task ledger, provider trace, memory, or operator state is required. The cockpit is an observation and routing surface: even its `--check` mode runs public structural checks, not Lean, and cannot promote a navigation result or workbench note into a theorem.

The underlying tour command returns a short card rather than a database dump. It derives the first result to inspect, its contribution class and boundary, the eight-problem map with its open count, and the next route commands from the committed projections. The full packet uses a registry-scaled budget: 18 kB of base context plus 2 kB per indexed problem, hence 34 kB for the present eight-problem registry. It expands the card into

a mathematical map, status counts, reader-specific contracts, and typed follow-up commands.

Five intent lenses cover the main transitions:

1. understand the mathematics through a selected programme and its claims;
2. locate any formal object through ordinary-language search followed by exhaustive inventory;
3. inspect exact formal dependencies through proof cones and paths;
4. begin a checked change through the session notary and focused builder;
5. audit the agent and release system through the publication architecture and release gate.

The route logic names no specimen theorem or module. Repository-specific mathematics enters through data: programme rows, claims, declaration records, and frontier propositions. This distinction matters. The mechanism is generic over the repository's projection schemas, while the current projection content is intentionally specific to the mathematics it describes.

Ordinary-language entry is likewise not an undocumented trigger list. The `--vocabulary` packet is an executable Rosetta surface: it publishes question operators, aliases, transparent expansions, typed route hints, an all-problem registry projected from `docs/problems.json`, and a route-discovery lexicon projected from `docs/claims.json::machine_readable_paper.entrypoints`. A unique exact authored term may take a constant-time fast path; ambiguous or unseen language falls through to visible vocabulary interpretation and ranked corpus search. Every search response records that interpretation, so an agent can inspect how its wording was translated instead of depending on phrase luck.

The packet also answers four different cold readers. A research mathematician gets programmes, claim statuses, and surviving open propositions. A formalisation engineer gets declaration coordinates, exact value references, and a route into a checked change. An AI-lab researcher gets coverage and authority distinctions. An independent contributor gets the no-build orientation, changed-cone build route, and public handoff check. Each reader sees the same facts through a different question contract rather than a separate hard-coded example.

4 From navigation to proof authority

The architecture treats navigation and proof as separate capabilities (Figure 2). A static query may nominate a theorem or dependency path. It cannot establish that a proposed application typechecks, that a proof closes a goal, or that an English interpretation is mathematically faithful.

The session notary records a typed move grammar in an append-only ledger: observations, conjectures, plans, interpretations, abandonments, probes, claims, and closure. Conjectures may declare a falsifier.

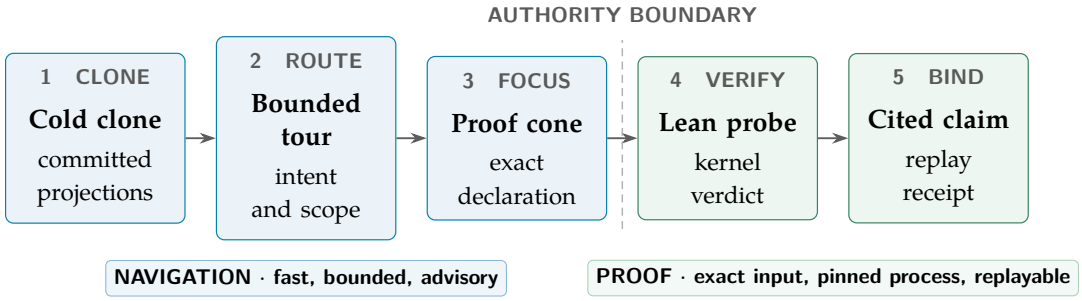


Figure 2: The shortest path from first contact to an authoritative receipt. The agent chooses the route; the notary derives the verdict from pinned Lean execution.

A probe is different. The notary stores the exact Lean input bytes, runs the pinned Lean process, and computes one of three verdicts from the process result: accepted, accepted with an admitted placeholder, or rejected. The agent cannot supply a verdict through the supported notary interface. A claim must cite an accepted probe receipt; the notary rejects weaker citations. Replay reruns every stored probe and compares the current result with the recorded verdict. The continuous-integration probe runner, used by sessions without a local build, gives a weaker receipt: its acceptance records that Lean compiled the file with no error and no `sorry`, that the file declares no axiom and that its printed axioms are the three standard ones, and it marks `exact_target_checked` false; the case study's probe bytes and kernel receipt are committed in its session ledger under `research/workbench/sessions/`.

Workflow-provenance systems already record what ran, when and where it ran, how tasks and data were connected, and who initiated them; they also distinguish a planned workflow from the trace of an executed one [12]. That is the closest established role for the notary's execution metadata. The notary adds a different contract: its ledger types an agent's conjectures, abandonments, probes, and claims, while the notary derives probe verdicts from the pinned Lean process. This interface rule does not prevent an actor with filesystem access from changing the implementation or stored record. The comparison does not show that the resulting record improves the agent's research choices.

This design deliberately leaves search policy open. LeanDojo couples an open programmatic Lean environment with extracted proof data, premise annotations, retrieval, and a theorem-proving benchmark [4]. Pantograph exposes tactic execution, proof states, metavariables, and data extraction through machine interfaces [5]. Those systems make machine-to-Lean interaction richer. The notary here addresses a complementary question: after an agent chooses its own policy, which parts of the resulting reasoning record are advisory, and which exact claims are grounded by replayable kernel acceptance?

5 Compilation after comprehension

A committed navigation projection is portable text; an `.olean` file is a toolchain- and platform-specific build product. A cold clone can therefore orient immediately but must either compile the selected formal target or restore a compatible cache before editing.

The focused build wrapper accepts an exact target, modules changed since a Git reference, or the stale cone derived from Lake traces. It orders the selected modules into dependency waves, limits parallel jobs, and finishes with a serialized Lake authority check. In one historical four-file addition, the planner selected 26 modules in four waves rather than both complete roots; in a warm two-root dogfood plan it selected no stale modules.

The exhaustive dependency-index validator is expensive for a different reason: it traverses the elaborated environment and exports direct references. After a full export matches the committed index, the validator writes an exact receipt below the ignored Lake cache. Its key includes every supported Lean source file, the toolchain and Lake locks, the exporter and builder inputs, the declaration atlas and generated manifest, the formal-source release slice of the claim registry, the dependency-extraction helper definitions, and the committed index bytes. An ordinary check reuses the receipt only while all inputs and output remain byte-identical; an explicit full-check mode bypasses it. A changed formal or exporter input therefore forces a fresh incremental build and full export, while documentation-only revisions and exact cache restores do not.

The receipt does not claim that the environment export itself has become incremental. When Lean source changes, Lake still recompiles only stale dependencies, but the graph exporter currently scans the supported environment again.

6 A replayed case study

The committed prospective session is a naturalistic use of the workbench, not a synthetic benchmark. The ledger contains six reasoning notes, one kernel-accepted probe, and two claims bound to that probe. It produced a binary carry-pivot normal form for numeral-adjacent boundary words and an exact divisor-incidence identity that sharpens a previously landed one-sided coefficient bound to equality. At the audited revision, replay reruns the stored probe and returns the same accepted verdict.

The receipt supports a narrow statement: the stored input is accepted by the pinned Lean environment, the cited claims point to that accepted probe, and the result survives replay at this revision. It does not establish that the agent's route was optimal, that every candidate was searched, that a bounded failure is impossible, or that the mathematics is new outside the repository history.

The navigation surface was dogfooded separately as four audiences. The research-mathematics route reaches the programme map and explicit frontier; the formalisation route reaches a source coordinate and exact proof cone; the AI-lab route exposes exhaustive-versus-selective coverage and receipt authority; the contributor route reaches a focused build and release check. Adversarial tests remove or distort each protected transition and require the cold-clone validator to reject the mutation. This evaluates the declared first-contact contract, not subjective comprehension.

7 Relation to prior work

Proof blueprints pair informal plans with named Lean declarations and author-supplied dependency links. `leanblueprint` stores a blueprint in TEX and its checker only confirms that each named declaration exists [1]; it neither infers those links nor checks that the informal and formal statements agree. LeanArchitect infers formal dependencies and proof-hole status and exports synchronised blueprint material [2]. The Carleson project shows a blueprint used as a large collaboration surface. Its public blueprint split the formalisation into about 180 claimable tasks, most corresponding to one blueprint lemma; formalisation feedback produced localized corrections, modifications, and extensions to the mathematical guide, including a few changes to the general setup and main theorems [3]. The present tour has a different first-contact role. It begins after the repository already contains a large corpus, distinguishes exhaustive inventory from authored interpretation, and routes a reader into exact dependency and proof-receipt tools.

LeanDojo and Pantograph provide stronger interaction substrates for learned or scripted theorem proving [4, 5]. This work does not propose a new proof-search policy. It makes the policy slot explicit and records enough of an agent-chosen trajectory to separate notes, nominations, rejected probes, and accepted claims.

Nor is its declaration graph or retrieval route novel in isolation. LeanGraph extracts typed elaborator-level edges across Lean projects, while a separate network study analyses Mathlib's multilayer declaration graph [6, 7]. LeanExplore combines semantic, lexical, and graph ranking behind Python and MCP interfaces, and LeanSearch v2 targets global premise sets through iterative sketch–retrieve–reflect [8, 9]. The narrower claim here is that one cold-clone contract composes source addresses, selective semantics, dependency cones, incremental validation, and authority receipts without requiring a hosted service or an initial Lean build.

Repository question answering provides a second comparison. Abedu, Khatoonabadi, and Shihab combine an LLM with a knowledge graph built from users, commits, issues, and files; their evaluation spans five open-source repositories and a 20-participant task study [13]. They explicitly leave code entities and program-analysis edges to future work. The tour here does not evaluate a conversational interface. It commits code-level Lean declarations, elaborated dependencies, authored mathematical interpretations, and claim-status boundaries so that a reader can inspect the route before an LLM explains it.

Large-library maintenance supplies the relevant compilation lesson. *Growing Mathlib* describes performance-aware library design, deprecation, semantic linters, benchmarks, review tooling, and explicit technical-debt management [10]. The changed-cone planner and receipt cache apply a smaller-scale version of that maintenance posture. The local measurements do not transfer Mathlib's scale or social evidence.

Lean Atlas narrows the declarations whose meaning can affect selected theorem statements and leaves semantic verification to people [11]. That is close to the authority distinction here: formal dependency can focus human inspection without deciding intended meaning. The cold-clone tour adds a front-door contract across exhaustive inventory, selective semantics, public claims, and agent receipts.

8 From a proof receipt to a public return

A proof receipt closes only the validation job. It does not say whether a claim record, paper, computation, open frontier, query route, or contributor surface has become stale. The clone therefore continues through an explicit sequence: invoke a bounded route with `mine-open-problem`; continue, suspend, or close the research run at its stated stopping condition; validate the stable delta; classify its downstream consumers with `propagate-research-consequences`; package the evidence and provenance with `erdos-research-return`; and prepare the proposed upstream change with `submit-pull-request`. Pushing a branch or opening the pull request still requires the contributor's authorisation.

The router also has to serve a request that names no target. The task "read the corpus and find something worth developing across the papers" matched no cue and fell through to the repository-explanation fallback, so an agent asked to think was handed a tour. The repair followed the method of this paper: the observed request became a routing fixture, a lane for open reading was added, and the `explore-the-corpus` skill now hands an emerged question to `mine-open-problem` and the sequence above. The same research instruction is published without a clone in the reading edition.

The sequence is deliberately independent of the agent harness. A contributor may use Codex, Claude Code, Cursor, another runner, ordinary scripts, or no agent. The shared contract is the starting commit, changed source, evidence, claim ceiling, surviving limitation, consequence dispositions, and stop or re-entry condition. A suspended run can therefore be resumed without being misreported as either failure or proof.

The starting commit also permits work from an older clone to survive current development. Git retains the common ancestor and hence the original delta. Maintainers first reproduce that delta in its stated context, then reconcile the reviewed substance with current main. Validation and consequence propagation are repeated on the integration delta. Material conflict resolution is recorded and credited as new work; the contributor's original authorship is preserved. The complete governance sequence appears in the companion contribution protocol, while the wider job map appears in the systems lifecycle.

9 Limits and transfer conditions

This is a case study of one repository at one audited revision. Its projection schemas and routing engine are reusable mechanisms, but another project must supply its own source roots, programme map, claim vocabulary, frontier statements, and authority owners. The paper gives no evidence that a new project can adopt the architecture cheaply or that agents using it outperform agents with an ordinary README.

The audit was author-operated. No external contributor had yet completed the tour, attempted the documented return path, or reported where the first-contact explanation failed. Cold-clone usability and maintainer burden are therefore open empirical questions, not properties established by the internal replay.

The repository also carries a source-frozen four-task reader pilot and a controller that checks material membership and captured trace accounting. Those artifacts make

an experiment more reproducible, but no reader run or independent grade accompanies them. All four tasks share one mathematical family, so they cannot support an estimate of transfer across problem families. The study's conditions and evidence limits are described in the companion systems paper's evaluation section.

Counts describe artifact coverage, not mathematical understanding. Direct dependency edges are not proof explanations. Authored semantic nodes may be wrong or incomplete. Maintainer-reviewed claims may also be wrong; automation preserves recorded relationships but does not judge unrestricted prose.

Receipted reasoning is auditable rather than necessarily good. A model may choose unproductive probes, omit a relevant source, or rationalise a dead end. Replay detects environment drift in stored probes but not every flaw in the surrounding interpretation. Context-blind historical experiments also cannot establish training-data blindness.

Finally, the first full project build and the first elaborated graph export remain material costs. The architecture moves them after comprehension, supports compatible cache restoration, and avoids repeating an exact export when nothing relevant changed. It does not eliminate the cost of validating a genuinely changed formal environment.

10 What the architecture guarantees

An agent-native formal repository needs to reveal, from a cold clone, what mathematics exists, what each navigation layer knows, what remains open, and which next action crosses into proof authority. The architecture here realises that sequence with committed projections, a public proof cockpit, a bounded intent tour, exact loaded-root dependency data, a session notary, focused builds, and cache-bound validation receipts.

The central design choice is separation. Comprehension comes before compilation; nomination comes before application; reasoning notes remain distinct from kernel verdicts; exhaustive routing remains distinct from selective interpretation; and a public claim remains distinct from the formal statement it describes. Those boundaries let a capable agent move quickly without making speed look like authority.

A How to reproduce the audit

The public repository commits the tour, route, projections, workbench session, and validators used in this paper. The shortest reproduction route has four bounded tasks.

1. Orient without compiling Lean. These commands inspect committed projections only:

```
python3 scripts/proof_cockpit.py --check
python3 scripts/query_corpus.py --tour --format card
python3 scripts/query_corpus.py --route agent_native_corpus_navigation
python3 scripts/query_corpus.py --search <ordinary-language-query>
python3 scripts/query_semantic.py inventory <candidate-name> --limit 1
python3 scripts/check_cold_clone_comprehension.py --quick
```

The cockpit check validates the public claim registry, cold-clone contract, and generated-orientation freshness; it does not run Lean. The full tour packet is obtained by omitting

the format flag. A declaration name returned by inventory can be expanded into a direct neighbourhood or bounded proof cone.

2. Check a proof. Beginning formal work enters the notary and then the focused builder:

```
python3 scripts/proof_workbench.py open --help
python3 scripts/lean_fast_build.py --jobs 2 --lake-staleness <target>
```

3. Check reuse of the dependency index. The ordinary check may reuse an exact cache receipt; the full check bypasses it:

```
python3 scripts/test_lean_dependency_index_cache.py
python3 scripts/build_lean_dependency_index.py --check
python3 scripts/build_lean_dependency_index.py --check --full-check
```

The first ordinary check after a cold clone may perform the full export; a matching restored Lake cache can carry the exact receipt. The full-check form always bypasses it.

4. Inspect the return lifecycle. The clone-local owners can be read without installing them globally:

```
cat skills/mine-open-problem/SKILL.md
cat skills/propagate-research-consequences/SKILL.md
cat skills/erdos-research-return/SKILL.md
cat skills/submit-pull-request/SKILL.md
python3 scripts/test_clone_skills.py
python3 scripts/test_contribution_entry.py
```

These commands inspect and test the route. They do not push a branch, open a pull request, accept a contribution, or alter mathematical status.

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